Automating the Classification of Economic Activities in Official Statistics: A Comparative Study of Neural Networks and Transformers

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Abstract: - This paper explores the process of automation of the classification of open-ended questions regarding the economic activities of enterprises, in official statistics. Neural networks (NN) and transformer-based models such as BERT are compared. This study's aim is the improvement of the accuracy and efficiency of the economic activities classification in alignment with the NACE classification system. The textual data from official statistics are processed beforehand. NN and BERT models are utilized to classify at the 2-digit and 4-digit levels. To assess and compare the effectiveness of these models, performance metrics, methods such as accuracy and F1-scores, are used. The results show the potential of transformer-based models to improve the process of automation codification of economic activity, by the reduction of manual work and increasing consistency in classifications. This research makes an essential contribution by exploiting the potential of the application of transformer models to domain-specific data such as the ones in official statistics, advancing the field of automatic text classification.

Key-Words: - text classification; machine learning; official statistics; transformers; neural networks, supervised learning.

JEL Classification: C45, C88

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1 Introduction

With the progressing of digital transformation, the automatic classification of text data originating from various sources has become crucial. As a field, official statistics require timely and qualitative processing of data. Text classification techniques are central to many tasks in natural language processing (NLP) [1], such as the classification of economic activities according to standard classification systems, like the NACE (Nomenclature of Economic Activities). Naturally, transforming text responses to open-ended questions has been conducted manually, which is tedious and can result in human error. It adds even more complexity to the task when we have larger data sets. A powerful alternative to manual codification is the replacement with machine learning, especially with neural networks and transformers.

Neural networks present great potential for text classification because they excel at learning complex patterns from data, [2]. More recently, transformers—deep learning architectures—have changed the landscape in natural language processing. They have enabled more sophisticated handling of context and semantics within text than before, [3].

By understanding the concept, as a result it has been taken one step further by transformer models such as BERT, which is able to comprehend the context of words in a sentence way better than any other traditional system. This is so as proposed in [4].

This paper attempts to showcase ways neural networks and transformers classify text responses against economic activities. By making use of the power of these deep models, we have worked to construct a solid solution that will efficiently code economic activities with high accuracy. The proposed method is not only likely to save manual effort but also to increase the efficiency and robustness of the classification process. This presentation will discuss how neural networks and transformers can enhance automation in text classification for official statistics through experiments and analyses conducted in this study, and it will also contribute to developing a more efficient and reliable methodology for processing economic data.

2 Literature Review

2.1 Overview of Text Classification

In the context of Natural Language Processing (NLP), one of the most useful tasks is Text Classification. It is the process of organizing text data into specific topics. The earlier approaches to text classification dealt with Naive Bayes, support vector machines (SVMs), and decision trees. These techniques are based on hand-crafted features such as term frequency – inverse document frequency (TF-IDF) and bag of words representations. [5]. None of these approaches provide good results for the complexity and nuances of text data.

With deep learning, the text classification dilemma has improved significantly. Text is now classified using several types of neural networks, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), shifting the focus to hierarchical features where neurons specialize in extracting specific features from the data, [6]. The advent of transformers has further advanced the capabilities of models to understand context and word relations in documents, [3]. Both BERT and its successors have excelled in text classification because of their multiple layers of encoders, which allows them to comprehend tremendous amounts of text and their contexts deeply, [4]. In high-level economic texts where deep semantic and contextual understanding is essential, transformers are very useful, [6].

In the area or field of economics and public statistics, the text classification technique has been applied for the analysis of open-ended questions in a survey, sectoral classification of news articles, and putting companies into categories based on their economic undertakings. These are all tasks that require an understanding of the economic context and the ability to manage considerable amounts of text, which indicates the need for powerful and scalable classification models, [7].

2.2 Machine Learning in Official Statistics

Driven by a broader availability of data and the need for more efficient processing techniques, the application of machine learning in official statistics has risen. It has been applied in a number of cases, such as missing data imputation, outlier detection, and classification of text survey responses, [7]. Particularly, there have been preliminary attempts to apply machine learning models for automating the codification of socio-economic responses – for example, in the classification of occupation titles or economic activities from text entries.

This task has been accomplished using traditional machine learning approaches like decision trees and SVMs, but these methods have high costs because of the extensive feature engineering that is required. They also do not transfer well across different datasets. More advanced models have been developed, such as deep learning CNNs and RNNs, which are expected to perform better for complex text data that do not have extensive ,manually crafted features, [8]. Unfortunately, these models are used in official statistics, and they always have a problem with the need for explanation and evaluation of quality according to international official statistics standards.

Recent advancements with transformers are addressing some of these challenges. Studies have demonstrated that transformer models can be fine-tuned for domain-specific applications, such as the classification of economic activities from open-ended survey responses, [9]. Moreover, their ability to pre-train on large datasets and fine-tune on smaller, domain-specific ones makes them particularly appropriate for the application in official statistics.

2.3 Gaps and Opportunities

Even with some advancements in using machine learning for proposal classification in official statistics, a number of gaps still remain. One of the gaps is the lack of particular understanding of the domain models that result from the context of economic activities. Most models an expert might consider using are trained on ‘the one size fits all approach’ and generic datasets, which makes them powerful but unfit for use in economic data, which encapsulates unique scripts and terminologies. Therefore, their utility is limited in official statistics, especially where classification accuracy and knowledge about the domain are highly needed.

Additionally, there are further proposals for the interpretability and transparency of these models for official statistics. For example, there must be consideration taken into the models utilized that perform well. With regard to the importance of official statistics, It is imperative to also have clarity into the rationale behind the models. The need for interpretability is often very crucial in the area of advanced AI usage, like neural networks and transformers, which are even more often referred to as ‘black box’ technologies.

With the gaps highlighted, this paper will seek to address these issues within the context of the classification of Economic activities on the basis of NACE classification, contributing towards an

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enhanced methodology of data processing in official statistics.

3 Methodology

In this paper, neural network and transformer-based models were used for economic activity classification based on the NACE classification system. The dataset used was from the Statistical Business Register. It was pre-processed, taking into consideration linguistic elements of the Albanian language. Suitable pre-processing of text, including stop word removal, text cleaning, and normalization, is critical to improving the accuracy of machine learning models, [10].

3.1 Data Collection and Preprocessing

3.1.1 Data Source

The dataset used in this paper comes from the Statistical Business Register. The description of economic activities in the Albanian language is used. The data was codified based on the NACE classification. Figure 1 shows the distribution of the data in NACE at the 4-digit level, while Figure 2 illustrates the distribution at the 2-digit level.

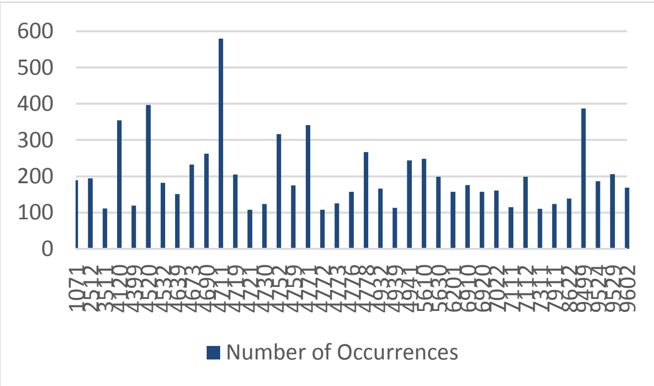

Fig. 1: Occurrences of each product code in 4-digit level

Source: Author's calculation

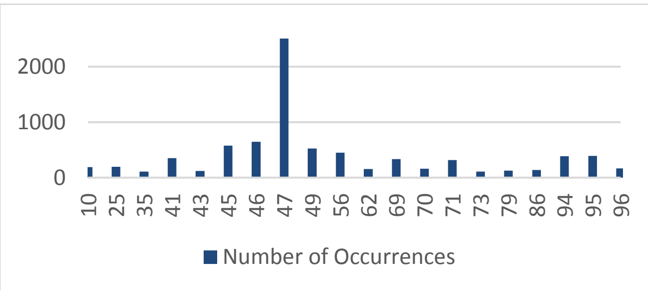

Fig. 2: Occurrences of each product code in 2-digit level

Source: Author's calculation

The initial preprocessing took care of missing values by removing records with incomplete 'narrative' or 'product' items and also resetting the index to maintain data integrity.

3.1.2 Text Cleaning and Normalization

The pre-processing task involved several key steps taking in consideration the Albanian language:

Stop word Removal: A set of Albanian stop words was recognized, which was used to remove from the text many words that do not contain any information. This step helps in focusing on the content that has meaning related to economic activities.

Text Cleaning: Text was cleared by replacing special characters with spaces and removing symbols not relevant to the classification. The cleaning process was also careful regarding the Albanian-specific letters such as 'ë' and 'ç'.

Lowercasing and Normalization: All text was lowercase. Numerical digits were also removed to reduce noise.

Length Calculation: Based on the number of words after pre-processing, the length of each narrative was calculated.

The cleaned and preprocessed text data was then prepared for input for the neural network and BERT models to make sure that the text was standardized and did not contain noise.

3.2 Feature Engineering

To prepare the data for the neural network model, tokenization and padding were used. A tokenizer was initialized to limit the vocabulary size. The text was converted then to lowercase. The tokenizer was fitted on the text descriptions from the 'narrative' field to generate a dictionary of unique words that exist in the text. Each narrative was then transformed into a sequence of integer tokens representing the words. To ensure that all text sequences had a consistent length, they were amplified or reduced. This is done to ensure that input sequences are of uniform size.

BERT tokenizer was used to process text data. The multilingual BERT tokenizer was initialized to handle the input texts. The tokenizer was also used to generate attention masks in order to help the model distinguish meaningful tokens and padding during training. Afterward, they were converted into tensors to make sure that the input data could be effectively processed by the BERT model during training and evaluation.

3.3 Model Selection

Chosen Models:

- **Neural Networks (NN):** A multi-layer neural network was implemented. It contained an embedding layer, which was initialized using pre-trainedpre-trained word embeddings, dense layers with ReLU activations. One softmax output layer was used to classify the text into NACE classifications.
- **BERT (Bidirectional Encoder Representations from Transformers):** BERT was fine-tuned on the NACE classification dataset, using its architecture to capture contextual relationships in the Albanian text provided. The output from BERT was fed into a dense layer to produce the final classification.

Both models were fine-tuned to improve their performance.

Regarding Neural Networks ,parameters such as learning rate, batch size, and number of hidden units were tuned. Dropout layers were included in the model to mitigate overfitting.

BERT fine-tuning was done by using a small learning rate with gradual warmup, early stopping, and learning rate scheduling to enhance the model's adaptability to the Albanian language and terms used in the dataset.

3.4 Experimental Setup

Training and Testing:

The dataset was split into training, validation, and test sets. This split was done by using an 80-10-10 ratio. Stratified sampling was used to maintain representative distributions of classes. Cross-validation was performed on the training data to tune the models and evaluate their robustness.

Evaluation Metrics:

The models were evaluated using:

- **Accuracy:** To measure the overall percentage of correctly classified instances.
- **Precision, Recall, and F1-Score:** To measure the performance, taking into consideration the impact of imbalanced class distributions.
- **Confusion Matrix:** To evaluate the models' performance on different NACE classes and help to identify areas of potential enhancement.

Cross-Validation and Resampling

In order to evaluate the performance, k-fold validation was used. The dataset was split into five subsets. Every one of these subsets was used for validation, while the others were used for training the model. Average metrics for all folds were calculated to evaluate model performance.

As a result of limited computational resources, bootstrap resampling was used for 10 iterations. 95% confidence intervals for precision, accuracy, recall, moreover F1-scores were also calculated.

3.5 Model Complexity and Training Time

In this study, the transformer model (BERT) and the neural network (NN) were trained and evaluated on the same dataset. It is very important to emphasize the fact that even though the transformer has a better performance than the neural networks when it comes to k-fold validation or bootstrapping, they require very high computational resources. In terms of training times, the NN model for the 4-digit was consistent, with an average of around 21 seconds per epoch. The NN model for the 2-digit presented a similar scenario. The average time for epochs was about 20 seconds. Both NN models were run on CPU and continued to be efficient in terms of speed during the training process. On the other hand, the transformer models presented longer training times. The 4-digit transformer took about 50 seconds per epoch, and the 2-digit transformer took around 49 seconds per epoch. For the transformer models, a GPU (T4) was used, which resulted in a significantly higher need for computational resources. The difference in training time shows that while transformers show better performance results, the resource intensity compared to the neural network models is notable.

4 Results

4.1 Performance of Transformer Models

The transformer models were evaluated for the 2-digit and 4-digit level NACE classification. Below are the results:

Transformer Model (2-Digit Classification):

- **Accuracy**: 0.84
- **Macro Avg**: Precision: 0.80, Recall: 0.80, F1-Score: 0.80
- **Weighted Avg**: Precision: 0.84, Recall: 0.84, F1-Score: 0.8

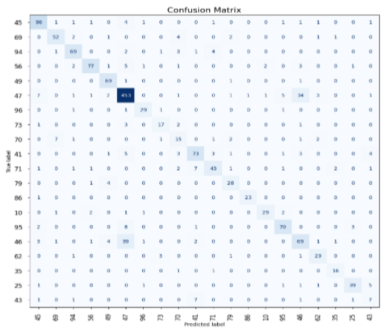

Fig. 3: Confusion Matrix for Transformer Model on 2-Digit Classification
Source: Author's calculation

As shown in Figure 3, the confusion matrix for the transformer model highlights a high performance for frequent classes such as class 47 with 453 correct classifications. The matrix shows that the model performs well for frequent classes. Nevertheless, there are frequent misclassifications for some classes, such as class 46 is often misclassified as 47 and the opposite. This is most probably happening due to context similarities in the text description of these classes. The same is true with classes 70 and 69, which commonly share the same words in description. Classes with low frequencies, such as 43 and 25, have poor performance in the classification. This is due to the very low number of training samples. Generally, misclassifications tohappen with similar or neighbor classes, so it is difficult for the model to differentiate them.

Fig. 4: Training and Validation Loss and Accuracy for Transformer Model on 2-Digit Classification
Source: Author's calculation

Figure 4 presents the training and validation loss and accuracy curves for the transformer model during the 2-digit classification. These curves demonstrate the model's efficient learning process: the loss decreases steadily, and the accuracy increases consistently over time, indicating that the transformer model is successfully learning from the data without significant overfitting.

Transformer Model (4-Digit Classification):

- **Accuracy**: 0.73
- **Macro Avg**: Precision: 0.73, Recall: 0.73, F1-Score: 0.73
- **Weighted Avg**: Precision: 0.73, Recall: 0.73, F1-Score: 0.73

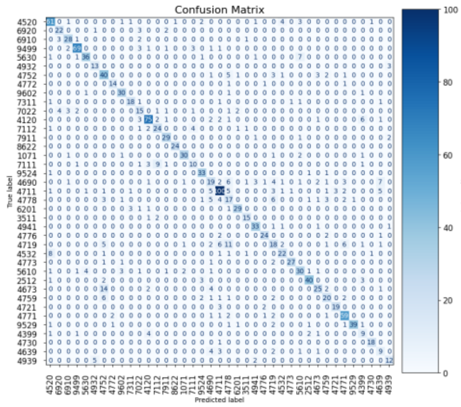

Fig. 5: Confusion Matrix for Transformer Model on 4-Digit Classification
Source: Author's calculation

The 4-digit classification task presents more complexity, as indicated by the performance drop for both models. In Figure 5, the confusion matrix for the transformer model shows that it continues to perform well, especially in the more frequent classes. High-frequency classes such as 4711 and 4120 have higher classification performance. Still, misclassifications occur with similar classes, such as class 4752 and 4763, which have similar descriptions. Low-frequency classes such as 4939 and 4639 are not performing well due to the limited number of records in these classes.

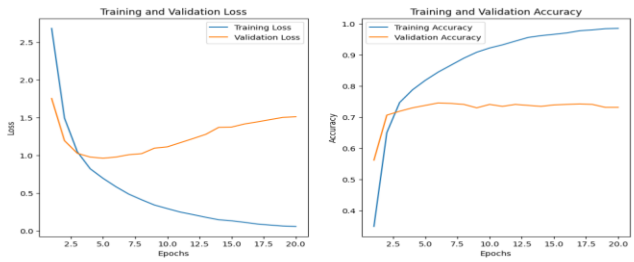

Fig. 6: Training and Validation Loss and Accuracy for Transformer on 4-Digit Classification
Source: Author's calculation

Figure 6 presents the loss and accuracy curves for the transformer model on the 4-digit classification. The curves demonstrate stable learning, with both training and validation accuracy gradually increasing while maintaining a close relationship. This indicates that the transformer model continues to generalize well, even in the more complex 4-digit classification task.

4.2 Performance of Neural Network Models

The NN models were evaluated for the 2-digit and 4-digit level NACE classification. Below are the results:

- **Neural Network Model (2-Digit Classification):**
  - **Accuracy**: 0.79
  - **Macro Avg:** Precision: 0.76, Recall: 0.76,
  - F1-Score: 0.76
  - **Weighted Avg**: Precision: 0.79, Recall: 0.79, F1-Score: 0.79

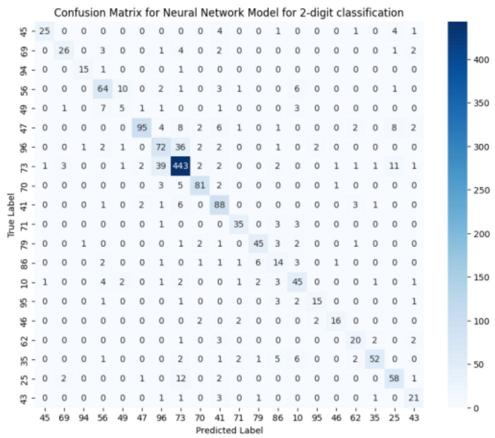

Fig. 7: Confusion Matrix for Neural Network on 2-Digit Classification
Source: Author's calculation

Similarly, Figure 7 provides the confusion matrix for the neural network model on the 2-digit classification. Classes such as 47 or 69, as also shown in the transformer model, are performing better. Neighbor classes suffer from confusion as a result of overlap in their descriptions. Additionally, low-frequency classes, such as 43 (21 correct classifications) and 46 (16 correct classifications), still have low classification performance. The neural network performs well in frequent classes. Its lower recall and F1 scores in rare classes show that it has more than the transformer model due to class imbalance.

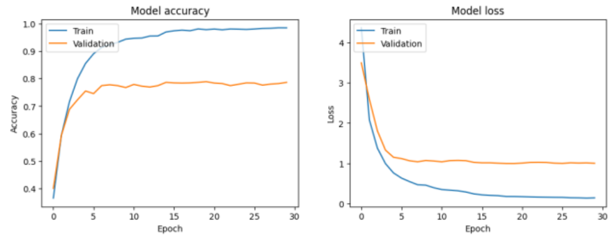

Fig. 8: Training and Validation Loss and Accuracy for Neural Network on 2-Digit Classification
Source: Author's calculation

Figure 8 shows the training and validation loss and accuracy curves for the neural network. Unlike the transformer model, the neural network exhibits a slower decrease in loss and a larger gap between training and validation accuracy. This gap indicates a tendency towards overfitting, as the model struggles to generalize well to unseen data, especially in the later stages of training.

For the 4-digit classification, the results were as follows:

- **Neural Network Model (4-Digit Classification)**:
  - **Accuracy**: 0.68
  - **Macro Avg**: Precision: 0.69, Recall: 0.69, F1-Score: 0.69
  - **Weighted Avg**: Precision: 0.68, Recall: 0.68, F1-Score: 0.68

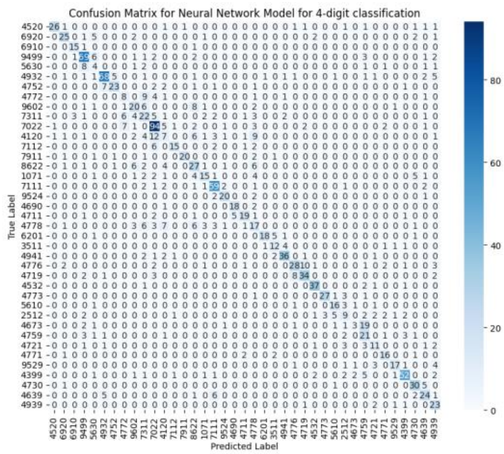

Fig. 9: Confusion Matrix for Neural Network on 4-Digit Classification
Source: Author's calculation

Figure 9 displays the confusion matrix for the neural network model on the 4-digit classification, where we see a higher drop in performance compared to the transformer model. The problem with similar classes and underrepresented classes here is more evident. The neural network particularly has difficulties with rare classes.

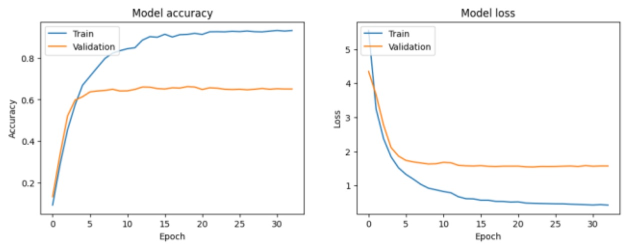

Fig. 10: Training and Validation Loss and Accuracy for Neural Network on 4-Digit Classification
Source: Author's calculation

In Figure 10, the training and validation loss and accuracy curves for the neural network show a larger gap between training and validation accuracy. The pattern in the validation loss suggests that the model is overfitting, especially in the later epochs. This is likely contributing to the neural network's weaker performance in classifying rare and complex categories. In the 4-digit classification, both models saw a drop-in performance compared to the 2-digit task. However, the transformer model once again outperformed the neural network with higher accuracy and F1 scores.

- **5-fold validation cross-validation**

For the 5-fold cross-validation at a 2-digit level, the confusion matrix is presented in Figure 11. The diagonal in this matrix is more visible, and the values on it are higher counts, which shows better accuracy than the 2-digit model without k-fold.

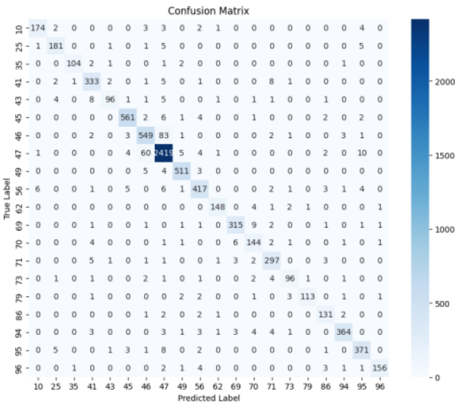

Fig. 11: Confusion Matrix for Neural Network on 2-Digit Classification
Source: Author's calculation

Figure 12 presents the feature's importance. It shows that the model is very much relying on Features 29, 28, and 27. Feature 29 is the most important, with a mean importance higher than 0.2. Features lower than 25 have a minimal contribution and are most probably redundant or irrelevant.

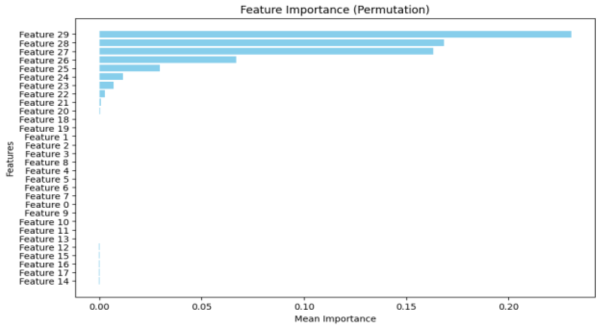

Fig. 12: Feature Importance for Neural Network on 2-Digit Classification
Source: Author's calculation

The mean residual is 0.2634, suggesting a minor positive bias, showing that the model tends to overestimate true class values. The standard deviation is 1.5001, indicating a reliable performance.

The residual distribution is presented in Figure 13. It shows that the model predicts correctly in the majority of cases. Occasional misclassifications are shown with a very low frequency.

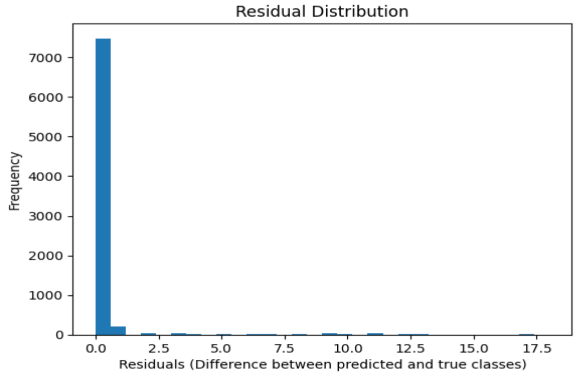

Fig. 13: Residual distribution for Neural Network on 2-Digit Classification
Source: Author's calculation

Bootstrapping was used to evaluate the calculated model accuracy. The 95% confidence interval was calculated as [0.9355, 0.9457]. It shows a narrow range, demonstrating consistent performance across resampling iterations.

For the 5-fold validation at the 4-digit level, the confusion matrix is presented in Figure 14. The matrix shows that the diagonal is stronger than in the 4-digital classification without cross-validation. Off-diagonal misclassifications are also lower. Aggregated results on folds show that the consistency is better.

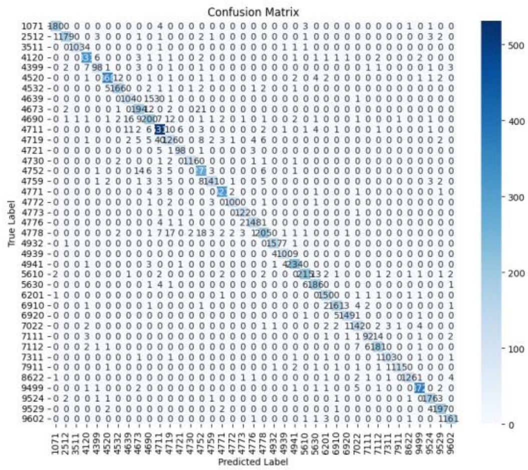

Fig. 14: Confusion Matrix for Neural Network on 4-Digit Classification
Source: Author's calculation

While Figure 15 presents the feature importance of the case of the 4-digit model with 5-folds. It shows that the model is relying on Features 29, 28, and 27. Feature 29 is the most important one. Features lower than 23 have minimal or no importance.

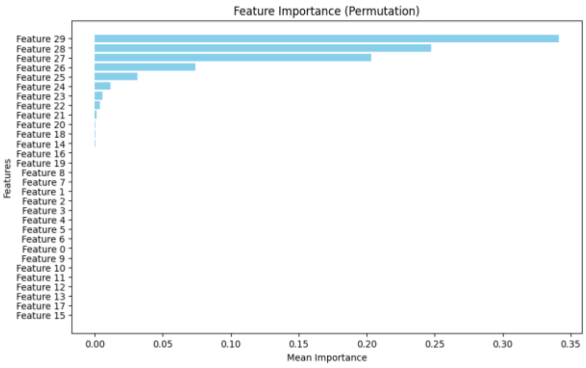

Fig. 15: Feature Importance for Neural Network on 4-Digit Classification
Source: Author's calculation

The mean residual is 0.7007, presenting a higher bias than the 2-digit model. The standard deviation is 3.2541, indicating a less reliable model.

The residual distribution is presented in Figure 16. The majority of residuals are concentrated at 0 or very close to it, showing that the model frequently predicts correctly. It also rarely makes big mistakes.

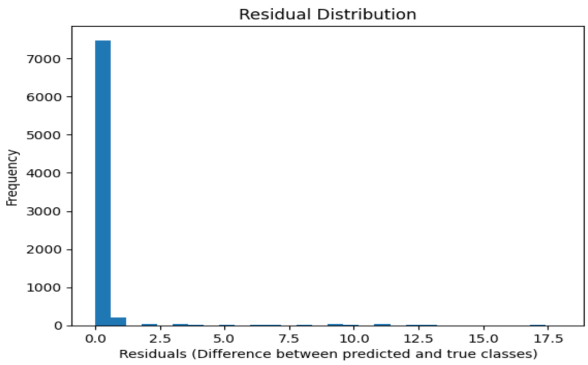

Fig. 16: Residual distribution for Neural Network on 4-Digit Classification
Source: Author's calculation

Bootstrapping was also used for the 4-digit level classification. The 95% confidence interval was calculated as [0.88909845 0.90217528]. It shows lower performance than the 2-digit classification, with a mean of 89.56%.

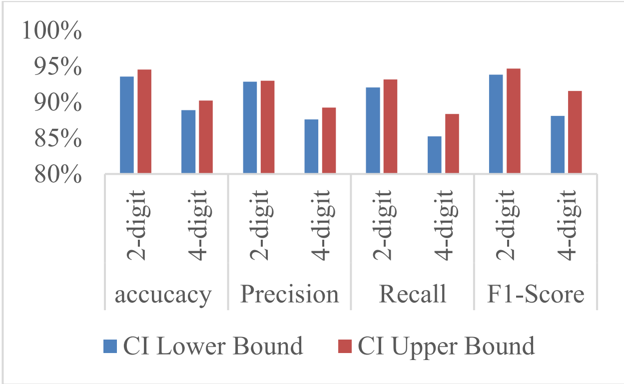

Fig. 17: Performance Metrics with 95% Confidence Intervals (2-Digit vs. 4-Digit Classification)
Source: Author's calculation

Figure 17 presents confidence intervals for all metrics: accuracy, precision, recall, and F1-score. For all metrics, the 2-digit classification shows a higher performance. All the lower and upper bounds are greater than the ones for the 4-digit classification. Confidence intervals for the 2-digit classification are between 93.55% and 94.67%, respectively, for accuracy and F1-score. The 4-digit classification has lower metrics. The confidence intervals for accuracy and F1-score are between 88.91% and 91.57%. The 2-digit classification has tighter intervals that indicate better consistency in model performance.

4.3 Narrative Length Analysis

The analysis of text description lengths shows a high level of variation. The average length of the text description is 4.08 words. The standard deviation is 1.95. At the 2-digit level, the shortest narrative is class 96 (average length 2.79 words), which impacts the classification performance as it sometimes suffers from sufficient context. Longer narratives, on the other hand, such as in the case of class 46 (5.28 words), provide more descriptive information. When it comes to the 4-digit level, 9602 is the class with the lowest length, 2.79, while 4690 and 4639 have the longest descriptions, respectively, 5.71 and 5.21 (in line with the 2-digit level).

4.4 Token Frequency Analysis

The most frequent tokens in the dataset contain words such as 'pakicë' (1207 occurrences – meaning retail) and 'tregti' (678 occurrences – meaning trade). For class 96, there are specific terms such as 'estetike' (28 occurrences – meaning aesthetic) and 'parukeri' (21 occurrences – meaning beauty salon), giving the context of beauty services. The same happens when detailing to the 4-digit level with class 9602: 'estetike' (28 occurrences – meaning aesthetic) and 'parukeri' (21 occurrences – meaning beauty salon), because class 96 has only 9602 as a representative in the 4-digit level.

4.5 Shared Tokens Between Classes

Class 96 shares a large number of tokens with other classes, such as 47 (58 tokens), where we can mention "services,"" products," and "coffee." It shares with class 46 39 tokens, showing a potential semantic overlap. A similar situation is with classes 95, 45, and 56. Class 56 and 49 share 60 tokens, explaining also misclassifications presented in the confusion matrixes.

4.6 Class-Level Results

At the class level, the transformer consistently performed better than the neural network, particularly for frequent classes. For example:

- Transformer F1-scores for Class 5 (2-digit level): 0.89 vs. Neural Network F1-score: 0.85
- Transformer F1-scores for Class 3 (4-digit level): 0.88 vs. Neural Network F1-score: 0.81

For rare classes, both models struggled, but the transformer model still held an edge in terms of precision and recall. For instance:

- Class 18 (4-digit level): Transformer's F1-score was 0.39, while the neural network achieved an F1-score of 0.33.

The performance comparison between the transformer and neural network models demonstrated that transformers consistently performed better than the neural networks, especially in handling imbalanced classes. It showed a better capacity capturing long-term dependencies and contextual relationships in complex datasets [3]. Moreover, the findings of this study showed that deep learning models, including transformers, are better suited for handling domain-specific text classification tasks, such as the classification of economic activities [10].

5 Discussion

The comparison between the transformer and neural network models indicates a clear advantage for the transformer, particularly in both the 2-digit and 4-digit classification tasks. The transformer achieved higher accuracy, precision, recall, and F1 scores across almost all metrics. This superior performance is likely due to the transformer's ability to comprehend more complex contextual relationships within the text data, which is particularly important in the hierarchical nature of the NACE classification.

In contrast, the neural network model, even though it performed well in general, it lagged in terms of handling complex or rare categories. The lower accuracy in the 4-digit classification suggests that the neural network struggles to distinguish between more granular categories, especially when dealing with imbalanced datasets.

K-fold cross-validation with stratified sampling was used to confirm the evaluation of model performance, as the dataset is imbalanced. The metrics were averaged across folds to check for variability. The cross-validation showed higher model stability. For example, the standard deviation of accuracy for the NN in 5-fold cross-validation was 1.5, indicating steady performance.

Shorter text descriptions, like in the case of class 96, give limited context, affecting, in this case, the performance of the models. Frequent tokens like "retail" may impact the classification in the direction of the overrepresented terms. The shared tokens between classes point to context similarities that impact classification between neighbors or similar classes.

The analysis highlights that shorter narratives, such as those for Product 96, result in reduced contextual richness, complicating classification. Moreover, frequent tokens like "retail" and shared tokens such as "services" exacerbate semantic overlap between products, leading to misclassifications.

As demonstrated in the confusion matrices and supported by the accuracy charts, both models were affected by class imbalance, especially in the 4-digit classification task. However, the transformer model shows better resilience, likely due to its architecture's ability to understand complex contextual relationships. The neural network's larger overfitting tendency, reflected in its loss and accuracy charts, suggests that future work on balancing techniques, such as SMOTE, could improve its performance a lot.

The transformer model, despite being more complex, appears to generalize better to unseen data in the NACE classification tasks. This complexity makes sure to capture finer differences between different classes at the 4-digit level, where the neural network shows lower performance. However, the transformer model also requires more computational resources and time for training, which is a limitation in practical applications with big datasets.

The transformers constantly outperform NNs across various metrics (accuracy, precision, recall, and F1 scores). This is largely due to their ability to understand long-range dependencies and context through mechanisms like self-attention, [3].

In terms of computational resources, transformers' superior performance comes at the cost of longer training times. As shown in the training and validation loss curves (Figure 4, Figure 6, Figure 8 and Figure 10), the transformer model stabilizes earlier and maintains better generalization.

Another aspect of transformers is their interpretability. Neural networks and transformers are often considered "black boxes" because it is difficult to understand how they make their classifications. This can be a possible problem to the adoption of official statistics, where decisions taken based on automated systems need to be transparent.

For transformers, one way to address this issue is through attention visualizations, which can show the words or phrases the model takes more into consideration when making the predictions. Future work should be focused on integrating attention visualization tools to increase the transparency of the BERT model's decision-making process. This would allow statistical agencies to better understand the model's behavior, especially in classification tasks such as the one regarding economic activity.

The models developed in this study have the potential to automate the classification of economic activities, reducing the manual effort and time needed. However, integrating these models into real-world workflows, such as those at INSTAT, needs additional work.

Deploying transformer models for large-scale classification tasks at INSTAT or other statistical agencies would require investment in infrastructure, such as access to GPUs or cloud-based solutions, to handle the computational demands.

6 Conclusion

This research compares transformer and neural network models for the automatic codification of economic activities using the NACE classification. The classification is done at 2- and 4-digit levels. Considering the results, it is indicated that the transformer model consistently performs better than the neural network. It achieves higher accuracy, precision, recall, and F1 scores in both classification tasks.

The advantage of the transformer model is especially shown in the fact that it is able to handle rare classes. Specifically, more complex categories at the 4-digit level. However, the class imbalance has an impact on both models, making rare categories have a lower performance.

The most frequent tokens, such as “retail” and “trade,” have the potential to bias the classifier. Domain-specific tokens such as “esthetical” and “salon” for class 96 are very important in correct classification but are underrepresented. Future work will use embedding that emphasizes these rare tokens.

The findings of this research show that while transformer models perform better than neural networks in classifying economic activities, particularly for detailed 4-digit categories, this performance comes with increased computational difficulties.

7 Future Work

Based on the models’ performance and the need to address their limitations, several directions for future work are proposed:

Explore strategies such as SMOTE, class weighting, or advanced data augmentation techniques to address class imbalance, with a particular focus on underrepresented categories. This could improve both transformer and neural network performance on rare classes.

Continue the work on the hyperparameter tuning, particularly for the neural network, in order to optimize their performance. Grid search or Bayesian optimization techniques may help fine-tune different parameters such as learning rate, batch size, and layer depth.

Explore additional domain-specific transformer models, such as FinBERT, [11]. Fine-tuning pre-trained models on domain-specific data can improve model accuracy, particularly for complex economic classifications at the 4-digit NACE level, [9]. Moreover, solutions that combine neural networks and transformers could offer a better solution, using the strengths of both approaches, [12].

In terms of insights on how decisions are made, tools that visualize attention weights in the case of transformers may help to increase transparency and provide explanations.

Future work will also focus on dealing with short narratives through augmentation and domain-specific embedding to handle token imbalance. Top features will be examined to exclude the ones with low importance and enhance efficiency.

Finally, deploying these models as pilots in a real-world scenario in the Statistical Office of Albania (INSTAT) would provide an important understanding of their practical application and limitations, considering both the infrastructure requirements and potential accuracy improvements in automating economic activity classification. Feedback from the practical use could guide further improvements and adjustments to the models.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the authors used Grammarly in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Conflict of Interest

The authors have no conflicts of interest to declare that are relevant to the content of this article.